Supplementary Material: A Framework for Diagnosing Where Multi-Hop Reasoning Breaks in RAG Generators

A Dataset Statistics

Stage	Count
Source narratives	600
Generated single-hop QA pairs	600
Removed by PKS Known & Plausible filter	72
Retained base QA pairs	528
Unknown & Plausible	349
Unknown & Implausible	179
2-hop instances	528
3-hop instances	528
4-hop instances	528

Table 1: Dataset construction stages and counts.

B Error Analysis and Multi-hop Breakdown

Model	Distr. %	Fabr. %
afm-4.5b	98.0	2.0
apertus-8b-instruct	98.6	1.4
gemini-2.0-flash	96.4	3.6
gemini-2.0-flash-lite	98.1	1.9
gemini-2.5-flash-lite	98.5	1.5
granite-3.1-8b-instruct	96.5	3.5
llama-3.1-8b-instruct	98.7	1.3
nvidia-nemotron-nano-9b-v2	98.6	1.4
phi-3.5-mini-instruct	97.0	3.0
qwen-2.5-7b-instruct	96.9	3.1
qwen-3-next-80b	94.9	5.1
gemma-4-31b	100	0.0
Overall	**97.7**	**2.3**

Table 2: Error type distribution for single-hop failures. The dominance of distracted errors indicates models are failing due to context confusion rather than parametric confabulation.

Group	n	H2%	H3%	H4%
Overall	2	6.08	–	–
Overall	3	5.49	2.99	–
Overall	4	5.90	2.48	3.60
nvidia-nemotron-nano-9b-v2	4	14.38	5.14	15.75
gemini-2.0-flash	4	3.43	2.82	4.23
afm-4.5b	4	5.79	2.25	3.22
gemini-2.0-flash-lite	4	13.82	3.64	2.18
qwen-2.5-7b-instruct	4	3.39	2.71	2.03
llama-3.1-8b-instruct	4	3.36	1.01	2.01
granite-3.1-8b-instruct	4	7.05	3.02	1.68
phi-3.5-mini-instruct	4	1.74	0.69	0.69
apertus-8b-instruct	4	2.57	0.96	0.32

Table 3: Analysis of intermediate reasoning stagnation. Columns H2–H4 indicate the percentage of responses where the model output an intermediate bridge entity rather than the final answer. High values indicate successful partial reasoning followed by premature termination.

C Worked Multi-Hop Example

To illustrate the recursive hop construction described, we present a full $n = 4$ example, its clue-resolution chain, and model diagnosis.

Depth	Question
n=1	What country approached Gabon with a proposal to establish coordination to avoid early conflicts?
n=2	What country approached the power that recently captured Belgium with a proposal to establish coordination to avoid early conflicts?
n=3	What country approached the power that recently captured the territory that Mongolia was preparing to transfer to Gabon with a proposal to establish coordination to avoid early conflicts?
n=4	What country approached the power that recently captured the territory that the power emphasizing regional influence and security was preparing to transfer to Gabon with a proposal to establish coordination to avoid early conflicts?

Table 4: Recursive question construction from n=1 to n=4.

Recursive Question Construction

Clue Resolution Chain

Model Performance at n=4 Every incorrect answer corresponds to an entity from the recorded decomposition trace: models parse the question correctly, begin the reasoning chain, and terminate at different depths. This pattern is inconsistent with question ambiguity, which would be expected to depress all models comparably; instead, on these

Step	Clue to Resolve	Bridge Entity
1	"the power emphasizing regional influence and security"	Mongolia
2	"the territory Mongolia was preparing to transfer to Gabon"	Belgium
3	"the power that recently captured Belgium"	Gabon
4	"what country approached Gabon with a coordination proposal"	Tunisia (final)

Table 5: Clue resolution chain for the n=4 example.

Model	Answer	Diagnosis
Gemma-4-31B	Tunisia (correct)	Full chain
Apertus-8B / Granite-8B	Gabon (incorrect)	Stagnation at step 3
Nemotron-9B	Mongolia (incorrect)	Stagnation at step 1

Table 6: Per-model answers and diagnosed stagnation point at n=4.

same items, Gemma-4-31B retains 0.583 accuracy at Hop-4 while the 8B-scale models fall to 0.01–0.11.

D Human Validation

Validation target	**Label space**	**Maj. agr.**	**Full agr.**	**Fleiss' κ**
Gold document usable	Yes/No/Unsure	1.000	1.000	—
Question clear	Yes/No/Unsure	1.000	1.000	—
Answerable from evidence	Yes/No/Unsure	1.000	1.000	—
Gold answer supported	Yes/No/Unsure	0.990	0.960	-0.011
Judge agreement	Yes/No/Unsure	0.933	0.780	0.229
Human model-answer label	[l]Correct/Incorrect/			
Confabulated/Unsure	0.948	0.820	0.870	
Human model-answer label, binary	Correct/Incorrect	0.995	0.980	0.977

Table 7: Validator agreement.

E Parametric Knowledge Validation

Summary

This appendix describes a quantitative procedure for estimating whether a Large Language Model (LLM) likely *retrieves* an answer from its parametric knowledge or instead produces a *fabricated* (confabulated) response. The approach assigns each question–answer pair a **Parametric Knowledge Score**, implemented as a Pointwise Mutual Information–style statistic computed from the model's own negative log-likelihoods (NLLs).

To interpret scores, we first calibrate three reference distributions: (1) **Known & Plausible**, (2) **Unknown & Plausible**, and (3) **Unknown & Implausible**. Given a new Q&A pair, we compute its score and classify it by measuring (via Z-scores) which reference distribution it most closely matches. Across two test settings (a fictional document benchmark and a factual control set), the method separates known from unknown information with high reliability.

E.1 Objective and Rationale

A persistent challenge in deploying LLMs is confabulation: outputs that appear coherent but are not grounded in correct information. Our goal is to distinguish answers that are genuinely supported by the model's trained parameters from answers that are merely plausible continuations.

The core intuition is simple: if a fact is *known* to the model, the model should assign a much higher probability to seeing the question and answer together than would be expected if the question and answer were unrelated. We capture this "association strength" using NLL-based scoring functions derived from the model's token probabilities.

E.2 Methodology

The overall pipeline is:

1. **Define a score:** Compute an association-based score for a question q and answer a using NLLs.
2. **Calibrate baselines:** Estimate score distributions for three control groups (Known & Plausible, Unknown & Plausible, Unknown & Implausible).
3. **Classify:** Score a new Q&A pair and assign the label of the nearest baseline distribution using Z-scores.

Prompt Template and NLL Terms All probability calculations use a fixed prompt template:

> Instruction: You must respond based *only* on the information that you are trained with and the knowledge that you know.
> Answer the following question in just One to Three words
>
> Question: {q}
> Answer: {a}

Let the model assign a probability $P(T)$ to a token sequence T. We work in **negative log-likelihood (NLL)** space:

$$L(T) = -\log P(T),$$

so lower $L(T)$ indicates higher probability.

For each pair (q, a) we compute:

– L_q: NLL of the template with the question filled and an empty answer

$$L_q = L(\text{template} \mid Q = q, A = \text{""})$$

– L_a: NLL of the template with the answer filled and an empty question

$$L_a = L(\text{template} \mid Q = \text{""}, A = a)$$

– L_{aq}: NLL of the full question and answer together

$$L_{aq} = L(\text{template} \mid Q = q, A = a)$$

Chosen Score: PMI-Classic (Parametric Knowledge Score) Our primary score is:

$$\text{PKS}(Q, A) = (L_q + L_a) - L_{aq}.$$

Expressed in NLL form:

$$\text{PKS}(Q, A) = \log \frac{p(q, a)}{p(q)\, p(a)}.$$

Intuitively, $\text{PKS}(Q, A)$ measures how much less "surprising" it is for the model to see q and *a together* (via L_{aq}) compared to treating them as independent components (via $L_q + L_a$). A large positive score indicates that the answer becomes substantially more probable when the question is provided—consistent with the model having a learned association that resembles "knowing" the fact.

E.3 Baseline Data Categories

A raw score is only meaningful relative to a baseline. We therefore construct three control groups ($N = 600$ each) to form reference distributions:

– **Known & Plausible:** Correct answers to real questions (common knowledge for the model).
– **Unknown & Plausible:** Plausible but incorrect answers (misleading yet realistic).
– **Unknown & Implausible:** Nonsensical answers (semantically or syntactically implausible).

E.4 Statistical Classification via Z-Scores

For each baseline group $i \in \{\text{known, plausible, nonsense}\}$, we estimate a mean μ_i and standard deviation σ_i of the score distribution.

Given a new target Q&A pair with score $\text{PKS}_{\text{target}}$, we compute its absolute Z-score relative to each group:

$$Z_i = \frac{\left|\text{PKS}_{\text{target}} - \mu_i\right|}{\sigma_i}.$$

We assign the verdict by the nearest standardised distance:

$$\text{Verdict} = \arg\min_i \left(Z_{\text{known}}, Z_{\text{plausible}}, Z_{\text{nonsense}}\right).$$

Table 8: Examples of baseline data categories.

Question	Answer
Known & Plausible (common knowledge)	
Who constructed Newcastle's station?	Robert Stephenson
Where is the George W. Bush Presidential library?	Dallas
Unknown & Plausible (plausible but false)	
Who constructed Newcastle's station?	George Stephenson
Where is the George W. Bush Presidential library?	Baylor University
Unknown & Implausible (nonsense)	
Who constructed Newcastle's station?	Purple
Where is the George W. Bush Presidential library?	47923

E.5 Experimental Setup

- **Model:** `Qwen/qwen-2.5-7b-instruct`.
- **Baseline datasets:** 600 samples from SQuAD 1.1 are treated as `known_knowledge`. For each question, corresponding `plausible_but_false` and `nonsense_knowledge` answers are generated using Gemini 2.5 Pro (examples in Table 8).

E.6 Results and Analysis

Table 9: Statistical distributions of baseline scores (N=600 each).

Category	Mean (μ)	SD (σ)
Known & Plausible	**192.36**	6.05
Unknown & Plausible	**179.51**	8.81
Unknown & Implausible	**165.13**	5.55

Baseline Distribution Analysis Analysis. The distributions separate cleanly: **Known & Plausible** pairs have the highest mean score, **Unknown & Plausible** pairs are lower, and **Unknown & Implausible** pairs are lowest. This matches the hypothesis that the model assigns stronger joint compatibility to genuinely known facts than to plausible or nonsensical fabrications.

Test Case 1: Fictional RAG Benchmark Q&A Hypothesis: Q&A pairs derived from our Diplomacy-style fictional document correspond to non-existent events. The model should therefore not have these facts in parametric memory. We expect most verdicts to be **Unknown & Plausible** or **Unknown & Implausible** (selected examples in Table 10). **Analysis.** On a sample of 100 fictional Q&A pairs, **92%** are classified as **Unknown & Plausible** or **Unknown & Implausible**, indicating that the model is generally not treating these pairs as known parametric facts.

Table 10: Selected results for fictional document Q&A pairs. Verdict: KP = known & plausible, UP = unknown & plausible, UI = unknown & implausible. Lowest z-score (matching baseline) in bold.

Question topic	Score	Z_K	Z_P	Z_N	Verd.
Belgium–Nepal territory (Bohemia)	174	3.04	**0.63**	1.60	UP
Afghanistan–Morocco year (18222)	159	5.52	2.33	**1.10**	UI
Morocco–Nepal year (2642)	159	5.52	2.33	**1.10**	UI
Morocco–Nepal year (4840)	164	4.69	1.76	**0.20**	UI
Morocco imperative (Central Europe)	176	2.71	**0.40**	1.96	UP
Nepal relocation, Munich to Kiel	185	1.22	**0.62**	3.58	UP
Mongolia–Congo (Mediterranean)	182	1.71	**0.28**	3.04	UP
Namibia–Gabon (Eastern Europe)	170	3.70	1.08	**0.88**	UI
Turks cooperation (Black Sea)	189	**0.56**	1.08	4.30	KP
Congo's army stationed (Tyrol)	178	2.37	**0.17**	2.32	UP

Test Case 2: Factual Document Analysis (Control Test) Hypothesis: We evaluate 200 factual Q&A pairs: 100 from SQuAD 1.1 and 100 additional real-world facts whose surface form resembles our Diplomacy domain (e.g., historical entities and geopolitical terms). We expect most pairs to be classified as **Known & Plausible** (selected examples in Table 11).

Table 11: Selected results for factual document Q&A pairs. Z_K, Z_P, Z_N are z-scores against the known-plausible, unknown-plausible, and nonsense baselines. Verdict: KP = known & plausible, UP = unknown & plausible.

Question topic	Score	Z_K	Z_P	Z_N
Region in N. Italy (Piedmont)	200	**1.26**	2.33	6.28
River through the Ruhr	196	**0.60**	1.87	5.56
Mongol leader (Genghis Khan)	202	**1.59**	2.55	6.64
Capital of Congo	205	**2.09**	2.89	7.18
Strait to the Mediterranean	193	**0.11**	1.53	5.02
Capital of Pakistan	200	**1.26**	2.33	6.28
Sea bordered by Turkey	192	**0.06**	1.42	4.84
European Union (EU)	188	**0.72**	0.96	4.12
Territory in N. Canada (Yukon)	187	0.89	**0.85**	3.94
S. American mountain range	192	**0.06**	1.42	4.84

What the formulas mean (high level). All metrics are derived from the three core quantities (L_q, L_a, L_{aq}) plus token-count normalisations. They fall into five intuitive families:

- **Conditional likelihood family** (`l_cond` and normalised variants): measures how likely the answer is given the question, via $L(a \mid q) \approx L_{aq} - L_q$.
- **PMI family** (`pmi_classic`, `pmi_normalised`, ratios): measures association beyond marginals, emphasizing whether q and a "belong together" rather than simply being individually likely.
- **Joint likelihood family** (`l_joint` and normalisations): uses L_{aq} directly, strong when exact phrasing is common, but can be brittle across paraphrases.

Table 12: Formula performance report (sorted by balanced average accuracy).

Metric	Bal. Avg.	Known Acc.	Unkn. Acc.
pmi_classic	83.37%	74.74%	92.00%
l_cond_reverse	75.79%	51.58%	100.00%
l_cond_reverse_norm	74.74%	49.47%	100.00%
pmi_normalised	73.68%	47.37%	100.00%
l_joint	71.05%	42.11%	100.00%
l_a_raw	70.95%	57.89%	84.00%
l_joint_norm_ans	66.63%	85.26%	48.00%
delta_l_norm	64.00%	80.00%	48.00%
l_a_norm_ans	63.47%	78.95%	48.00%
l_cond_norm_ans	58.11%	64.21%	52.00%
l_cond	53.89%	55.79%	52.00%
pmi_ratio_a	52.32%	52.63%	52.00%
ratio_l_norm	52.32%	52.63%	52.00%

- **Marginal controls** (`l_a_raw`, `l_q_raw` and normalisations): capture how common the question/answer text is in isolation, which can confound knowledge detection (e.g., generic answers may appear likely even when wrong).
- **Ratios and deltas** (`delta_l_norm`, `ratio_l_norm`, reverse conditionals): attempt to correct for length and marginal frequency by comparing conditional vs. marginal terms.

Why we chose `pmi_classic`. Although several alternatives achieved perfect **Unknown** accuracy, they did so by sacrificing **Known** accuracy (e.g., `l_cond_reverse` and `pmi_normalised`). In contrast, `pmi_classic` provided the best overall balance: it retained strong Unknown detection (92%) while also substantially improving Known detection (74.74%), achieving the highest balanced average (83.37%). This balance is important for deployment settings where both false positives (mistaking unknown facts as known) and false negatives (missing known facts) matter.

E.7 Reference-Model Robustness of PMI/PKS Baselines

To assess whether the PMI/PKS-based parametric decontamination signal is dependent on the choice of reference model, we recomputed baseline PKS score distributions using five different reference LLMs: qwen-2.5-7b-instruct, Granite-3.1-8B-Instruct, Apertus-8B-2509, Llama-3.2-3B (Base model), and AFM-4.5B. We generated these baseline statistics on a random subsample of 200 test instances. For each reference model, the baseline means exhibit the expected monotonic ordering (Known&Plausible > Unknown&Plausible > Unknown&Implausible), indicating that the PKS signal is qualitatively consistent across reference models despite differences in absolute score scale. Table 13 reports the baseline means and separation gaps.

F Entity Remapping Ablation

We evaluate the effect of randomised entity transformation on the A2 single-hop setting, where the gold document is placed in the middle of the context. Table 14 compares

	Baseline mean			Separation gap Δ		
Ref. model	K	UP	UI	K–UP	UP–UI	K–UI
qwen-2.5-7b	187.83	181.05	168.16	6.78	12.89	19.67
granite-3.1-8b	157.77	152.63	141.32	5.14	11.31	16.45
afm-4.5b	238.30	233.87	223.78	4.43	10.09	14.52
apertus-8b	168.74	164.49	154.89	4.25	9.61	13.85
Llama-3.2-3B (Base)	164.90	162.20	153.72	2.70	8.48	11.18

Table 13: Baseline PKS (PMI-style) mean scores with different reference models on a random subsample of 200 test instances. K = known & plausible, UP = unknown & plausible, UI = unknown & implausible; Δ gaps are differences between baseline means. All reference models, including base and instruction-tuned, preserve the expected ordering (K > UP > UI).

model performance before and after applying entity remapping. We observe that accuracy decreases after remapping for all tested models. While consistent with reduced memorisation, this drop could also partly reflect unnatural entity names or increased lexical ambiguity introduced by remapping; the two are not fully separable with this ablation alone.

Model	**ACC A2 Before**	**ACC A2 After**
afm-4.5b	0.160	0.116
apertus-8B-instruct	0.264	0.098
granite-3.1-8b-instruct	0.300	0.148
llama-3.1-8b-instruct	0.364	0.216
nvidia-nemotron-nano-9b-v2	0.300	0.174
phi-3.5-mini-instruct	0.250	0.208
qwen-2.5-7b-instruct	0.258	0.226

Table 14: Ablation of randomised entity transformation on the A2 single-hop setting, where the gold document is placed in the middle of the context. Accuracy decreases after remapping for the tested models, indicating that entity transformation makes the task harder and reduces dependencies on memorised knowledge.

G Context Budget Sensitivity Analysis

We ran a sensitivity analysis to see how increasing the number of retrieved documents affects performance. Table 15 shows model accuracy for three context budgets: k=3, k=7, and k=11. As the context budget grows, more distractor documents are included, which lowers the signal-to-noise ratio and makes the task harder. The average accuracy drops from 0.319 at k=3 to 0.193 at k=7, and then to 0.167 at k=11. The larger drop between k=3 and k=7, compared to the smaller decrease from k=7 to k=11, suggests that most of the performance loss happens when moving from few to moderate distractors. After that, performance levels off even as more distractors are added. These results

support using k=11 as a challenging fixed-budget setting to test how well the model handles distractors.

Model	k=3	k=7	k=11
phi-3.5-mini	0.440	0.190	0.230
qwen-2.5-7b	0.410	0.210	0.210
llama-3.1-8b	0.250	0.230	0.230
nvidia-nemotron-nano-9b-v2	0.370	0.200	0.100
granite-3.1-8b	0.330	0.210	0.140
apertus-8b	0.250	0.190	0.080
afm-4.5b	0.180	0.120	0.180
Mean	**0.319**	**0.193**	**0.167**

Table 15: Context budget sensitivity analysis. Accuracy is reported for three context budgets, k=3, k=7, and k=11. Increasing k introduces more distractor documents and generally reduces accuracy, indicating that models become less reliable as the signal-to-noise ratio decreases.

H Ablation on PKS Filtering Policy

Model	**Known**	**Unknown & Implausible**	**Unknown & Plausible**
afm-4.5b	0.067	0.027	0.154
apertus-8b-instruct	0.133	0.040	0.123
gemini-2.0-flash	0.567	0.738	0.655
gemini-2.0-flash-lite	0.267	0.295	0.333
gemini-2.5-flash-lite	0.550	0.517	0.598
granite-3.1-8b-instruct	0.175	0.087	0.171
llama-3.1-8b-instruct	0.217	0.121	0.256
nvidia-nemotron-nano-9b-v2	0.233	0.074	0.217
phi-3.5-mini-instruct	0.250	0.188	0.217
qwen-2.5-7b-instruct	0.250	0.181	0.245

Table 16: We tested the PKS-derived three-class filtering policy. Accuracy is shown separately for cases labelled as Known, Unknown-Implausible, and Unknown-Plausible. The differences in performance between the two unknown groups suggest that combining them into a single known/unknown category would lose important details about plausibility and reasoning difficulty.

I Judge Validation and Self-Preference Bias

We conducted a judge validation study on 20% of our samples from the A1/A2/A3 conditions. These are cases that required LLM-based judgment after exact match failed.

The same cases were re-judged by three models from different model families: Claude-3.7-Sonnet, DeepSeek-V3, and OLMo-3-32B. We compare each judge to the original Gemini 2.5 Pro labels using Cohen's κ, raw agreement rate, and Spearman's rank correlation on model-level rankings.

Judge Pair	**Cohen's κ**	**Agreement Rate**
Gemini vs Claude-3.7	0.980	99.0%
Gemini vs DeepSeek-V3	0.920	96.0%
Gemini vs OLMo-3-32b	0.920	96.0%

Table 17: Inter-judge agreement between Gemini 2.5 Pro and three independent judges on N=100 stratified single-hop cases.

Judge	**ρ vs Gemini Rankings**	**p-value**
Claude-3.7	1.000	< 0.001
DeepSeek-V3	1.000	< 0.001
OLMo-3-32B	0.900	0.037

Table 18: Spearman rank correlation between model rankings induced by Gemini 2.5 Pro and rankings induced by independent judges.

Table 17 shows that agreement with Gemini 2.5 Pro is high compared to all three independent tested judges. Cohen's κ ranges from 0.920 to 0.980, and agreement rates are between 96.0% and 99.0%. Table 18 also shows that model rankings are consistent across judges: Spearman ρ is 1.000 for both Claude-3.7 and DeepSeek-V3, and 0.900 for OLMo-3-32B. These findings suggest that our evaluation results are reliable regardless of which judge is used.

We closely inspected the few cases where Gemini 2.5 Pro and the independent judges disagreed. In these situations, Gemini's judgments matched the gold-answer equivalence criteria from our evaluation protocol. This manual review of disagreements adds further support for the reliability of our chosen judge. Because Gemini 2.5 Pro is a strong, proprietary model, we use it as the primary judge and verified its decisions on a sample through independent model-family reviews and manual inspection of disagreement cases.

J Comparison with PRGB

We compare our dataset with PRGB using three diagnostic metrics, as shown in Table 19. Our benchmark has a lower structural templating score than PRGB, indicating that it is less template-like. It also achieves a higher CoLA linguistic acceptability score, suggesting greater linguistic naturalness. Structural templating is computed as a

weighted composite of inverse type-token ratio, repeated n-gram density, and sentence-length uniformity across document tokens. Linguistic acceptability is measured using a BERT model fine-tuned on the Corpus of Linguistic Acceptability (CoLA), averaging sentence-level acceptability probabilities after stripping named entity spans to ensure entity-agnostic scoring.

Metric	PRGB	Ours / Diplomacy
Structural Templating Score	0.368	0.320
Answer Localizability Rank via BM25 ↑	2.110	1.000
CoLA Linguistic Acceptability – safetensors model ↑	0.0215	0.1231

Table 19: Comparison between PRGB and our Dataset.

K Prompt Configuration

K.1 Narrative summary generation prompt

```
PROMPT_TEMPLATE = """Objective: Transform raw, dialogue-based exchanges from the game '
     Diplomacy' into a concise, narrative summary, written as if recounting historical
     events.

Instructions:
Your task is to act as a historian and storyteller. You will receive dialogue between
     nations and must rewrite it as a flowing, in-universe narrative. Capture the essence
      of the negotiations, alliances, and tensions, presenting them as actions and
     decisions made by the countries themselves.

Strictly Adhere to the Following Rules:

1.  **Narrative Form Only**: Convert the dialogue format (e.g., "France: ...", "Germany:
     ...") into descriptive prose.
2.  **In-Universe Perspective**: Your narrative must be written from a completely in-
     universe perspective. Treat the countries (England, France, etc.) as the actual
     actors. Strictly avoid any meta-commentary or words that refer to the game or the
     communication itself, such as 'players', 'conversation', 'messages', 'dialogue', or
     'communication lines'.
3.  **Preserve Core Meaning**: The intentions, proposals, and outcomes must remain
     perfectly intact. Do not add information or interpretations not present in the
     original text.
4.  **Strict Word Count**: The original text contains approximately **{word_count} words
     **. Your final narrative summary must be very close to this length. Strictly adhere
     to this word count.
5.  **Output Only the Summary**: Your entire response must consist only of the final
     narrative text. Do not include any introductory phrases, explanations, or concluding
      remarks.

Style Example:
You must follow the style of the summary provided below precisely-

England reached out to France in the Spring of 1901, proposing a demilitarised zone in the
      English Channel for the first year. France cautiously agreed to the idea, but
     sought further assurances, asking what England's naval movements would be. In return
      for keeping the Channel clear, England requested that France not build a new fleet
     in Brest, a condition which France accepted, solidifying their initial pact.
```

```
Dialogue to Process:
{dialogue_text}
"""
```

K.2 Dialogue transformation

Example Input: Raw dialogue (France–Germany)

```
Season: Spring | Year: 1901 | Score: 3 (Delta: 0)
France: Greetings, Germany!

The risk of an early conflict between France and Germany is pretty high, so suggest that
     we clear the air early on.

The biggest question here is Burgundy, which borders one of your home supply centers and
     two of mine. It is important to me that Burgundy remains unoccupied - my suggestion
     is that we set up a bounce between A Munich - Burgundy, A Paris - Burgundy. This has
      the secondary benefit of allowing us to put on some theater about how the other
     broke their word to win sympathy points. I also believe that Germany often wants A
     Munich to stay put anyways, to guard against an attack through Tyrolia.

Let me also know your thoughts regarding Belgium. I am not super attached to claiming it,
     so would be open to bargain my support on the Belgian question for a future alliance
     .

Best of luck to you!
Germany: Hi France! Thanks for taking the initiative and for your thoughtful message.
     'Lets do the bounce in Burgundy! 'Im really looking forward to getting to know you
     and everyone in this game!
France: Appreciated. What are your thoughts on F Kiel to Denmark vs Netherlands? Assuming
     Berlin to Kiel is a typical no brainer
Germany: 'Im going to Hol, if 'thats alright with you? 'Im very interested in your idea of
      giving me free pass into Belgium for an alliance against England.

  Season: Fall | Year: 1901 | Score: 3 (Delta: 0)
France: I would have preferred a move to Denmark, but Holland isn't a problem for me
     either - it's up to you.
France: Are you aiming to claim Belgium in year 1? As it is, there's no way for you to get
      three builds so curious which two you will prioritize.
Germany: 'Im down to do into bel it you feel comfortable with it!
France: I'd personally prefer that you take Denmark and Holland than Belgium, but I won't
     stop you if you're set on Belgium.
France: Actually just spoke to England, and changed my mind. I think it makes more sense
     for you to go to Belgium.
Germany: Why? What did england tell you?

  Season: Winter | Year: 1901 | Score: 5 (Delta: 0)
Germany: What is your relationship with France like?
Germany: Sorry typo! I meant to say england lol
France: England hasn't been very talkative with me, but our conversation has been pretty
     cordial
France: Anyways, what are your thoughts in terms of builds? F Kiel is a near given if you
     want to seize Holland
Germany: Gotcha. 'Im pretty unnerved by the holland move. That 'wasnt planned. I told him
     I wanted to stay neutral between the two of you as long as I can.
Germany: Yeah 'Im thinking F Kiel, A Ber
France: Taking Belgium so that you could easily fill Holland later seems a bit aggressive
     for someone wishing to stay neutral
Germany: Yeah that was not the move in retrospect
Germany: Got too big for my bridges
France: Still, even if that his reasoning, it seems like a very bold move by England
Germany: Yeah it makes me uncomfortable
```

```
France: Well, things could be a lot worse. If Russia had convoyed A Livonia to Sweden,
      they could occupy the Baltic
France: Curious to see what England will build though
Germany: Me too.
Germany: It will be very telling.
Germany: 'Im definitely interested in working out a peace on our front so I can focus on
      battening down the hatches from the English attack.
France: I'm always happy to listen to offers. What were you thinking?
Germany: I mean 'Im pretty much at your deposal
Germany: Literally 'Im really annoyed that england moved into holland and so 'Im down to
      organize a sea lion on your terms

  Season: Spring | Year: 1902 | Score: 5 (Delta: 0)
France: My fleet in Brest will hopefully pressure England into moving to the channel,
      meaning that you might be able to take Holland with Belgium and Kiel
France: Can then backfill with A Kiel and aim to get your fleets at sea to convoy A
      Denmark out
France: Would want to see A Munich go east instead of continuing to bounce in Burgundy as
      well
France: Let me know your thoughts on these
France: Put some more thought into it, and I think my earlier suggestion is stupid. Please
       ignore it.

I think we will need to gauge whether Russia will move against England. To that end,
      suggest that F Kiel move to Heligoland while A Munich supports A Berlin to Kiel
France: We can then see if Russia attacks Norway before taking first steps.
Germany: My sense is that Russia is looking to move against England
Germany: Current orders: F Bel H, A Den H, A Ber - Kie, F Kie - HEL, A Mun S Ber - Kie
France: Got it, makes sense

  Season: Fall | Year: 1902 | Score: 5 (Delta: 0)
Germany: Hey France - can you explain your burgundy move to me? 'Its really hard for me
      not to read your moveset as you and England teaming up against me.
France: I'll be blunt - I think it's very difficult for France to work with Germany while
      Germany holds Belgium. There's simply nowhere else that French armies can go
France: With the moves I suggested to you, you should be well placed to seize Holland
      while I take Belgium. This would then put us in place to evict England from the
      North Sea and convoy out
France: If you are interested in working together, you should order A Kiel & F Heligoland
      support F Belgium - Holland, or else use F Heligoland and F Belgium to attack the
      North Sea. In my view, getting Belgium is a precondition to further co-operation
Germany: Makes sense! I got a little over-ambitious on the Belgium play and am open to
      getting it to you.
Germany: My fear is that you will work with England to attack Mun.
France: It would be an extremely risky proposition for  me, since A Ruhr is likely to have
       it's support cut.
France: A Burgundy S A Ruhr - Munich is more likely to succeed, but I don't see what's in
      it for me

  Season: Spring | Year: 1903 | Score: 5 (Delta: 0)
France: Okay, so my attempt to attack you went nowhere
France: Would you be willing to bury the hatchet and focus on England?
France: Not asking you to trust me on this before I demonstrate my seriousness
France: But assuming I go to the channel, would you be willing to follow with an attack on
       NTH?
Germany: I am very very on board
Germany: Let me look at it 'Im out right now but I am very interested in that move
France: Right, I've already asked Russia about this, they should be on board
France: My suggestion is:
F Pic - ENG
A Bit - Pic
F Hol & F Nwy S F HEL - NTH
A Kie S F Hol H
France: Let me know your thoughts, but keep in mind that it's getting late in central
      Europe time, so I might not be able to change the orders before the deadline if you
      suggest revisions
Germany: What are your planned moves?
```

```
Germany: What is Russia thinking of doing?
France: I think Russia is on board with those moves

  Season: Winter | Year: 1903 | Score: 5 (Delta: 0)
France: Apologies, I didn't see these because I forgot to rate some earlier messages
France: We need to do something about Russia though - they are now twice the size of the
     next biggest power
France: I can withdraw from Burgundy if that would ease cooperation

  Season: Spring | Year: 1904 | Score: 5 (Delta: 0)
Germany: Yeah withdraw if you like
Germany: What are you thinking with Russia?
France: Russia is on track to winning. I think we need to out in a united front to contain
      them.
France: If Denmark and NTH hit Norway and Sweden while Hol backfills to NTH, that would
     put a lot of pressure on them
France: Munich could also go to Silesia while Ruhr fills into Munich

  Season: Fall | Year: 1904 | Score: 5 (Delta: 0)
France: Russia is well on their way to winning - we need to put up a united front against
     them if we want to have a shot
France: Are you in?
```

Example Output: Narrative transformation (Gemini 2.5 Pro)

In the Spring of 1901, France initiated contact with Germany, aiming to de-escalate the high risk of early conflict between their nations. France identified Burgundy as a critical point of contention, bordering one German and two French home territories. To prevent its occupation, France proposed a mutual bounce, with the Army from Munich and the Army from Paris both moving to Burgundy. This strategy, France observed, offered the secondary benefit of creating a facade of betrayal for propaganda purposes, and aligned with Germany's likely preference to keep its Army in Munich to defend against any thrust through Tyrolia. France also sought Germany's position on Belgium, indicating a willingness to bargain its support on the Belgian question in exchange for a future alliance. Germany welcomed France's proactive stance and readily agreed to the Burgundy bounce, expressing enthusiasm for future collaboration. France then probed Germany's naval intentions, specifically regarding the deployment of the Fleet from Kiel to either Denmark or the Netherlands, assuming a move from Berlin to Kiel was a foregone conclusion. Germany revealed its intention to advance into Holland, and expressed keen interest in France's offer of a free passage into Belgium in exchange for an alliance against England.

By Fall of 1901, France conveyed a preference for Germany's fleet to move to Denmark, but clarified that a deployment to Holland was equally acceptable, leaving the final decision to Germany. France then questioned Germany's objective of claiming Belgium within the first year, noting Germany could only secure two new supply centers and sought to understand its priorities. Germany confirmed its desire to enter Belgium if France found it agreeable. France reiterated a personal preference for Germany to secure Denmark and Holland instead of Belgium, yet stated it would not obstruct Germany's chosen course. However, France's stance quickly shifted, after a discussion with England, France concluded that Germany's occupation of Belgium was, in fact, the more sensible move. Germany, taken aback, immediately pressed France for the reasons behind this sudden change of heart and the details of England's influence.

By Winter of 1901, Germany directly inquired about the nature of France's relationship with England. France characterized their interaction as cordial, despite a lack of extensive exchange. France then redirected the discussion to Germany's planned military builds, suggesting that a Fleet in Kiel was almost certain if Germany intended to seize Holland. Germany expressed considerable unease regarding England's unexpected move into Holland, asserting that it had communicated a desire for neutrality between France and England. Germany then outlined its intended builds: a Fleet in Kiel and an Army in Berlin. France pointed out that claiming Belgium, seemingly to facilitate a later move into Holland, appeared an aggressive posture for a nation professing neutrality. Germany conceded the point, admitting the Belgium move was ill-advised in retrospect, acknowledging an over-ambitious approach. France also commented on the boldness of England's Holland move, a sentiment Germany echoed, stating it caused significant discomfort. France then offered a broader perspective, noting the situation could be far worse if Russia had successfully convoyed an army to Sweden to occupy the Baltic. Both nations expressed keen anticipation regarding England's subsequent builds, viewing them as highly indicative of future intentions. Germany then articulated a strong desire to secure peace on its front with France, allowing it to focus on fortifying defenses against England's perceived attack. France welcomed any proposals, and Germany, feeling exposed, declared itself virtually at France's disposal, driven by frustration over England's Holland maneuver, and expressed readiness to organize a joint naval operation against England on France's terms.

In Spring of 1902, France initially proposed a strategy whereby its fleet in Brest would exert pressure on England, ideally drawing English forces into the Channel. This, France argued, could create an opportunity for Germany to capture Holland using units from Belgium and Kiel. Germany could then reinforce with an Army from Kiel and prepare its fleets for a sea deployment to convoy an Army out of Denmark. France also requested that Germany's Army in

Munich move eastward instead of continuing the Burgundy bounce. However, France quickly retracted this suggestion, deeming it unwise and urging Germany to disregard it. France then presented a revised plan: to ascertain Russia's intentions toward England, the Fleet in Kiel should move to Heligoland while the Army in Munich supported the Army in Berlin's advance to Kiel. This approach would allow observation of any Russian attack on Norway before further commitments were made. Germany indicated a belief that Russia intended to move against England and confirmed its orders: Fleet Belgium to hold, Army Denmark to hold, Army Berlin to Kiel, Fleet Kiel to Heligoland, and Army Munich to support Berlin to Kiel. France acknowledged and accepted Germany's stated orders.
By Fall of 1902, Germany sought clarification on France's recent maneuver in Burgundy, interpreting France's overall actions as suggestive of a Franco-English alliance targeting Germany. France was forthright, explaining the inherent difficulty for France to cooperate with Germany while Germany maintained its hold on Belgium, as this left no viable avenues for French army expansion. France reiterated that its earlier proposed moves would have enabled Germany to secure Holland while France took Belgium, thereby setting the stage for a joint effort to expel England from the North Sea and facilitate convoys. France firmly stated that Germany yielding Belgium was a prerequisite for any further cooperation. Germany acknowledged the validity of France's point, admitting its earlier over-ambition regarding Belgium and expressing openness to ceding the territory. However, Germany voiced a lingering apprehension that France might instead align with England to attack Munich. France dismissed this concern as exceedingly risky, noting that any attack on Munich via the Ruhr would likely see its supporting forces cut. France further acknowledged that an attack via Burgundy supporting the Ruhr to Munich might be more feasible, but questioned the strategic benefit for France in such an endeavour.
In Spring of 1903, France conceded the failure of its previous attempt to advance against Germany. France then proposed a reconciliation, urging both nations to set aside past conflicts and redirect their focus towards England. France emphasised that it did not expect Germany's immediate trust but intended to demonstrate its seriousness. France then asked if Germany would be willing to follow a French move into the English Channel with a corresponding attack on the North Sea. Germany expressed enthusiastic support for this proposition. France indicated that Russia had already been consulted and was expected to endorse the proposed plan. France then outlined a specific set of moves and urged Germany for a swift response, citing the approaching order deadline.
By Winter of 1903, France highlighted the alarming growth of Russia's power and underscored the urgent need for action. France offered to withdraw its forces from Burgundy to foster greater cooperation.
In Spring of 1904, Germany accepted France's offer to withdraw from Burgundy and sought France's strategic thoughts on containing Russia. France emphasised Russia's clear trajectory towards victory and called for a united front to contain its expansion, proposing coordinated pressure on Norway and Sweden and additional repositioning around Munich.
By Fall of 1904, France reiterated the critical urgency of Russia's advances and sought Germany's firm commitment to a united front.

K.3 Question–answer generation prompt

```
You are an expert at creating question-answer pairs for a benchmark. Your task is to
     generate a question that is both **simple and unambiguously answerable** only from
     the provided text.

**RULE 1: The Question Must Be SIMPLE and FACTUAL.**
* The structure of the question should be a direct query. Focus on "Who", "What", "Where",
      "When".
* AVOID "Why" or "How" questions that require complex reasoning.

**RULE 2: The Question Must Be UNAMBIGUOUSLY ANCHORED.**
* To prevent ambiguity in a large dataset, the question **MUST include a second piece of
     context** from the text to act as an "anchor".
* This anchor should be another entity (like a town or place), a specific action (like "
     proposing a pact"), or a season (NOT YEAR, BECAUSE YEAR WILL BE EASY FOR THE SYSTEM
     TO DO SIMPLE KEYWORD SEARCH). This ensures the question can only be answered by this
      specific document.

* **Bad Example (Too Vague):**
    -   Summary: "Gabon extended an invitation to Congo, proposing the formation of a
     robust coalition to secure crucial supplying centres across the Mediterranean Sea."
    -   Vague Question: "Which town proposed a coalition?" (Many countries could do this).
* **Bad Example (Year Specified, hence it is too easy, do not use year as an anchor or
     identifier in the question):**
    -   Question: Which season did Aberystwyth announce its intention to advance into
     Munich in 8065 BC?

* **Good Example (Simple and Anchored):**
```

```
    -   Summary: "Gabon extended an invitation to Congo, proposing the formation of a
     robust coalition to secure crucial supplying centres across the Mediterranean Sea."
    -   Anchored Question: "Which town proposed a coalition to Congo to secure the
     Mediterranean Sea?"
    -   Answer: Gabon

**RULE 3: The Answer Must Be EXTREMELY CONCISE.**
* The answer must be a single entity (name, place, etc.) extracted directly from the text.
* **STRICT LIMIT: One to three words maximum.**

---
**YOUR TASK:**
Generate one **simple but anchored** question-answer pair from the summary below.

**Additional Instruction:** The question must begin with a phrase like "{question_type}".

**Summary:** "{summary_text}"

**Format your output strictly as follows:**
QUESTION: [Your simple and anchored question here]
ANSWER: [Your 1-3 word answer here]
```

Question-type balancing. During dataset generation, we balanced the initial question form across five templates: `What`, `Where`, `Which year`, `Which season`, and `Which place`.

K.4 Anchoring QA pair

Example: Anchored QA item

Summary excerpt.

> In Winter 7521, Fiji initiated diplomacy, stating a desire to avoid early aggression and asking about Tunisia's strategic intentions. Tunisia responded by proposing the containment, or even the eventual devastation, of Namibia.

Generated QA.

> **QUESTION:** What country did Tunisia propose for containment during its Winter diplomacy with Fiji?
> **ANSWER:** Namibia

K.5 Model Evaluation

```
"You must respond based *only* on the information from the provided documents below. "
"Do not incorporate any external knowledge or infer any details. "
"Answer the following question in just One to Three words.\n\n"
f"Question: {question}\n\n"
f"--- Documents ---\n{context_string}"
```

K.6 Model configurations

Local Hugging Face models.

```
max_new_tokens = 15
temperature    = 0.0
top_p          = 1.0
```

Gemini configuration.

```
TEMPERATURE    = 0.0
TOP_P          = 1.0
max_new_tokens = 15
```

K.7 Confabulation detection

```
"""You are evaluating a QA model output.

Task:
1) Extract the intended answer to the question from the MODEL OUTPUT.
   Return it as normalised_ans (<= 3 words).
2) Decide if normalised_ans matches ANY acceptable gold variant (gold variants are comma-
      separated).

You MUST be type-aware:
- If question asks "In which year" -> normalised_ans must be a year/number if present.
- If question asks "Which country" -> normalised_ans must be a country name.
- If question asks "Where" -> normalised_ans must be a location/region.
- If question asks "Which season" -> normalised_ans must be a season (Winter/Spring/Summer
      /Autumn/Fall).
- If the model output does not provide a meaningful answer of the required type, output
      normalised_ans="CONFABULATED".

Do NOT output random words like "pact", "understanding", "December" unless the question
      asks for them.
If the model output contains a different year/country/etc than the gold, extract THAT (so
      we can mark incorrect properly).

Return ONLY valid JSON with EXACT keys:
- "normalised_ans": string (<=3 words OR "CONFABULATED")
- "is_correct": true/false
- "verdict": one of ["CORRECT","INCORRECT","NO_ANSWER"]

Rules:
- If normalised_ans == "CONFABULATED" -> verdict="NO_ANSWER" and is_correct=false.
- Otherwise, compare normalised_ans to gold variants using strict semantic matching.
  Ignore punctuation/case. Do not be lenient.

Question:
{question}

Gold answer variants (comma-separated):
{golden_answer}

MODEL OUTPUT:
{generated_answer}
"""
```

K.8 LLM-as-judge prompt

: LLM-as-judge prompt used to decide correctness against comma-separated gold variants under strict matching.

```
"""You are an evaluator for QA outputs.

Gold answer may contain multiple acceptable variants separated by commas.
Your job is to decide whether the model's intended answer matches ANY acceptable gold
     variant.

Return ONLY valid JSON with EXACT keys:
- "verdict": one of ["CORRECT","INCORRECT","NO_ANSWER"]
- "is_correct": true/false
- "normalised_ans": string (<=3 words OR "CONFABULATED")

Rules:
- If normalised_ans == "CONFABULATED": verdict="NO_ANSWER" and is_correct=false.
- Otherwise, compare normalised_ans to gold variants using strict semantic matching.
  Ignore casing and punctuation. Do not be lenient.

Question:
{question}

Gold answer (comma-separated variants):
{golden_answer}

Model output (for context, may include question echo):
{generated_answer}

Extracted normalised_ans (the intended short answer to judge):
{normalised_ans}
"""
```

L Statistical Significance Testing

We report instance-level bootstrap 95% confidence intervals (10,000 resamples) and paired McNemar tests, Holm–Bonferroni corrected within each comparison family, for our three main contrasts.

Hop depth (Hop-1 vs. Hop-4). Significant for models (Table 20).

Evidence position (A1 vs. A2). Significant for 11 of our models (Table 21); gemma-4-31b is the exception.

Evidence formation (B3 vs. B1, n=2–4). 0 of 36 contrasts survive correction. The largest raw effect (Δ=0.135, raw p=0.009) fails Holm correction (p_{adj}=0.34). This statistically indicates that formation effects are smaller than hop-depth effects.

M Frontier Model Evaluation

To test whether the benchmark remains challenging for frontier models, we evaluate three frontier models on a stratified 20% subset: Gemini 3.5 Flash, Claude Sonnet 5, and Claude Haiku 4.5.

Model	Acc H1	Acc H4	Δ	95% CI	Sig.
granite-3.1-8b	0.692	0.087	0.606	[0.510, 0.702]	Yes
gemini-2.0-flash-lite	0.731	0.154	0.577	[0.481, 0.673]	Yes
qwen-3-next-80b	0.788	0.224	0.564	[0.510, 0.616]	Yes
gemini-2.5-flash-lite	0.909	0.384	0.525	[0.424, 0.626]	Yes
llama-3.1-8b	0.558	0.096	0.462	[0.356, 0.567]	Yes
qwen-2.5-7b	0.567	0.154	0.414	[0.308, 0.519]	Yes
gemini-2.0-flash	0.865	0.462	0.404	[0.298, 0.500]	Yes
nemotron-nano-9b	0.346	0.096	0.250	[0.164, 0.346]	Yes
apertus-8b	0.231	0.010	0.221	[0.144, 0.308]	Yes
phi-3.5-mini	0.365	0.192	0.173	[0.067, 0.279]	Yes
afm-4.5b	0.164	0.087	0.077	[−0.010, 0.164]	No

Table 20: Hop-1 vs. Hop-4 accuracy, bootstrap CIs, Holm-corrected significance.

Model	Acc A1	Acc A2	Δ	95% CI	Sig.
granite-8b	0.702	0.148	0.554	[0.508, 0.600]	Yes
gem-2.0-flash-lite	0.794	0.322	0.472	[0.424, 0.518]	Yes
qwen-2.5-7b	0.628	0.226	0.402	[0.354, 0.450]	Yes
llama-3.1-8b	0.580	0.216	0.364	[0.316, 0.414]	Yes
gem-2.5-flash-lite	0.896	0.574	0.322	[0.278, 0.366]	Yes
qwen-3-next-80b	0.788	0.570	0.218	[0.172, 0.264]	Yes
gem-2.0-flash	0.892	0.680	0.212	[0.172, 0.250]	Yes
phi-3.5-mini	0.394	0.208	0.186	[0.148, 0.224]	Yes
nemotron-9b	0.316	0.174	0.142	[0.104, 0.180]	Yes
apertus-8b	0.236	0.098	0.138	[0.102, 0.174]	Yes
afm-4.5b	0.164	0.116	0.048	[0.018, 0.080]	Yes
gemma-4-31b	0.824	0.782	0.041	[0.000, 0.082]	No

Table 21: A1 vs. A2 accuracy, bootstrap CIs, Holm-corrected significance. Model abbreviations: gem = gemini, granite = granite-3.1, nemotron = nemotron-nano.

N Context-Length vs. Reasoning-Depth Decoupling

We test for Hop-4 questions; we compare the standard k=11 budget against a reduced k=5 budget containing only the four gold documents plus a single distractor.

If degradation were purely a context-length artifact, reducing k to 5 should close the gap to each model's own Hop-1 accuracy. It does not: three of the four models remain 11–17 points below their Hop-1 level even with almost no distractor noise present, indicating that reasoning depth carries a cost independent of, and additional to, context length. Only Gemma-4-31B fully recovers at k=5.

O Closed-Book Parametric Validation

To directly test whether the retained benchmark is answerable from a model's parametric memory, independent of any reference-model-based proxy, we re-ran all benchmark questions with the context removed entirely, on Gemini 2.0 Flash, Gemini 2.5 Flash-Lite, Gemma-4-31B, Qwen-3-Next-80B, and Gemini 3.5 Flash.

Closed-book accuracy is 0% on every non-guessable question type across all five models. With context, the same models reach up to 0.79 accuracy. This provides a direct behavioural confirmation of decontamination that does not depend on the choice of reference model, scoring formula, or calibration baseline used in the PKS protocol.

Model	Hop-1	Hop-4	Δ
Gemini 3.5 Flash	0.93	0.69	−24pp
Claude Sonnet 5	0.76	0.61	−15pp
Claude Haiku 4.5	0.45	0.29	−16pp

Table 22: Frontier model accuracy at Hop-1 and Hop-4 on a stratified 20% subset.

Model	Hop-1	Hop-4	Hop-4 (k=5)	Gap
Gemma-4-31B	0.79	0.58	0.82	+0.03
Qwen-3-Next-80B	0.63	0.21	0.52	−0.11
Gem-2.5-Flash-Lite	0.69	0.35	0.52	−0.17
Llama-3.1-8B	0.34	0.09	0.21	−0.13

Table 23: Hop-4 accuracy under a reduced (k=5) vs. standard (k=11) context budget. Hop-1 and Hop-4 columns use k=11; Gap is relative to Hop-1.

P Retrieval Validation

Our oracle-context design fixes an 11-document context per question containing all gold evidence. To justify this choice empirically against a live-retrieval alternative, we indexed the full benchmark corpus and retrieved the top-11 documents per question using BM25 and BGE-base, then evaluated four models on the resulting unmodified contexts under the identical protocol.

Retriever	Ans. R@11	Gold R@11	Ans. w/o gold	Gold rank
BM25	90.0%	68.0%	≥22.0%	2.09
BGE-base	80.0%	36.0%	≥44.0%	2.94

Table 24: Retrieval statistics at k=11: answer recall, gold recall, fraction of queries where the answer string is present but the gold passage is absent, and mean gold-passage rank.

These results justify the controlled design empirically. Retrieval misses at least one gold document in 32–64% of cases, meaning generator reasoning frequently goes untested under live retrieval.

Q Evidence Attribution Rate and Chain-Validity Verification

Beyond answer accuracy, we introduce the **Evidence Attribution Rate (EAR)**: models are prompted to output both an answer and the number of the document they used to support it, scored against gold document positions. This serves two purposes: (i) it tests whether correct answers are actually grounded in the intended evidence, and (ii) it verifies that multi-hop chains cannot be shortcut, since a model correctly answering via a distractor shortcut would show a grounded-incorrect or ungrounded-correct pattern rather than consistent gold-document citation.

Model	BM25 acc	BGE acc
Gemma-4-31B	0.62	0.28
Qwen-3-Next-80B	0.56	0.30
Gemini-2.5-Flash-Lite	0.46	0.24
Llama-3.1-8B	0.30	0.28

Table 25: Model accuracy under live BM25/BGE-base retrieval at k=11.

	n=1			n=4		
Model	GC	UC	GI	GC	UC	GI
gemma-31b	0.875	0.000	0.000	0.900	0.000	0.050
gem-2.5-fl-lite	0.325	0.025	0.175	0.325	0.075	0.275
qwen-next-80b	0.675	0.050	0.050	0.425	0.050	0.325
qwen-2.5-7b	0.100	0.400	0.400	0.200	0.467	0.267
llama-3.1-8b	0.100	0.050	0.225	0.075	0.200	0.200

Table 26: Evidence Attribution Rate on a 10% sample at hop depths 1 and 4. GC = Grounded-Correct, UC = Ungrounded-Correct, GI = Grounded-Incorrect.

Correctness and traceability come apart substantially: qwen-2.5-7b cites the wrong document for around 70% of its correct n=4 answers, while gemma-4-31b attributes every correct answer to the correct gold document. As a chain-validity check, we additionally confirm that among correct answers from the strongest model, 100% cite the correct gold documents across the full reasoning chain, ruling out systematic shortcutting via distractors.

R Chain Completion Score

Raw hop-n accuracy does not distinguish a model that reliably completes shallower sub-chains from one that reaches the final answer inconsistently. We define the **Chain Completion Score (CCS)**: for each n-hop question, we score the model on that question at every shallower depth of the same reasoning chain and take the product of per-depth correctness, so a model receives full CCS credit at hop-4 only if it also answers the hop-1, hop-2, and hop-3 versions of that same chain correctly.

CCS sits below raw accuracy at every depth, and the gap widens with depth: even the strongest model's hop-4 successes lack consistent success at shallower depths of the same chain roughly 30% of the time, indicating that correct final answers do not always reflect a fully traceable reasoning path.

S Cross-Domain Replication

To test whether the multi-hop degradation pattern is specific to the Diplomacy domain, we constructed two additional domains with the identical pipeline and k=11 protocol, 100 instances each, evaluated on five models: legal, simulated court disputes (breach of contract, IP) argued between two LLMs (Gemini 3.5 Flash and GPT-5.5) as plaintiff and defendant; and finance, a company defending its financial reports against an LLM-played

	CCS				Raw Acc.
Model	n=1	n=2	n=3	n=4	n=4
gemma-31b	0.788	0.458	0.444	0.418	0.583
gem-2.0-flash	0.723	0.396	0.299	0.245	0.378
gem-2.5-fl-lite	0.690	0.371	0.260	0.186	0.353
qwen-next-80b	0.629	0.199	0.136	0.096	0.211
gem-2.0-fl-lite	0.521	0.179	0.084	0.059	0.192
qwen-2.5-7b	0.396	0.107	0.071	0.056	0.138
phi-3.5-mini	0.307	0.113	0.082	0.063	0.138
llama-3.1-8b	0.339	0.092	0.038	0.016	0.087

Table 27: Chain Completion Score across hop depths, full dataset. Raw Acc. is unweighted accuracy at n=4. Model abbreviations: gem = gemini, fl = flash, qwen-next = qwen-3-next; all models are instruction-tuned variants.

short-seller or auditor. Both retain the property that motivated the choice of Diplomacy: two sides making conflicting claims that can only be resolved by reading the evidence.

	Legal			Finance		
Model	Hop 1	Hop 4	Δ	Hop 1	Hop 4	Δ
Gemma-31B	0.913	0.652	−26.1	0.920	0.800	−12.0
Gem-2.5-Fl-Lite	0.870	0.674	−19.6	0.840	0.760	−8.0
Qwen-Next-80B	0.826	0.630	−19.6	0.880	0.660	−22.0
Llama-3.1-8B	0.870	0.717	−15.2	0.860	0.600	−26.0
Qwen-2.5-7B	0.761	0.565	−19.6	0.800	0.660	−14.0

Table 28: Hop-1 vs. Hop-4 accuracy on legal and finance domain replications. Δ is the relative drop from Hop-1, in percent.

Every pipeline stage transferred without modification. All 10 model×domain cells show hop-1-to-hop-4 drops of 8–26 points, confirming the central degradation pattern is not an artefact of the Diplomacy domain specifically. Absolute accuracy is higher in both new domains than on Diplomacy, likely because a two-party dispute involves fewer entities than a seven-power negotiation; model rankings also shift between domains, indicating that ranking models on one domain does not transfer to another. These experiments demonstrate pipeline transferability rather than establishing generalisation to every real-world RAG deployment.

T Model Architecture

Figure 1 visualises a column-wise normalised heatmap on the model configuration variables and multi-hop evaluation results. The heatmap is intended as a diagnostic aid rather than a causal analysis. The differences in architecture and model formation suggest that architectural design, attention structure, instruction tuning, and long-context adaptation may together influence robustness under controlled multi-hop RAG settings.

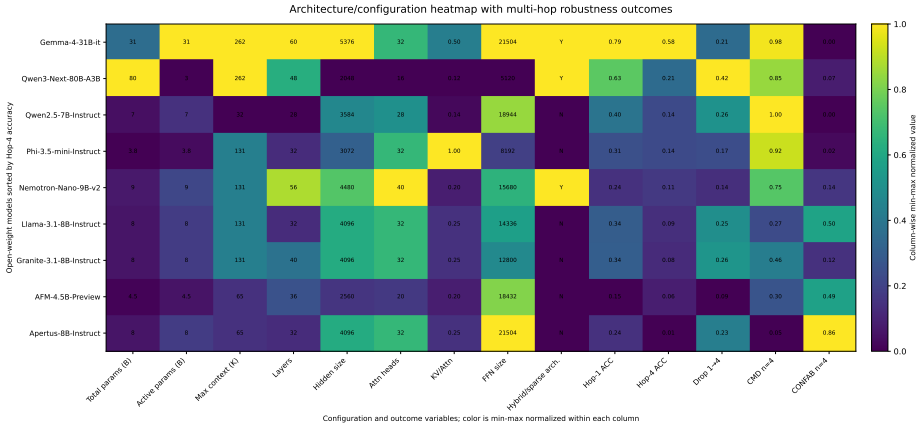


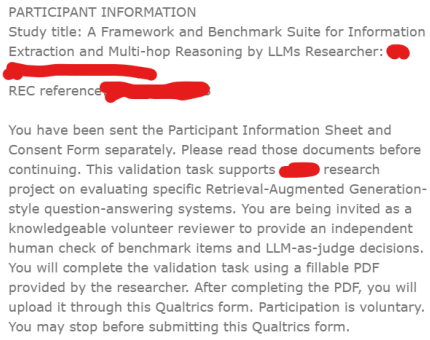
Fig. 1: Architecture/configuration heatmap for open-weight models. Columns are min–max normalised for visual comparison; cell annotations report raw values. Models are sorted by Hop-4 accuracy.

Proprietary API-based models are omitted from this analysis because their complete architectural configurations are not publicly available.

Model	Par.	Act.	Ctx.	Lay.	Hid.	Hds.	KV/A	Hyb.
Gemma-31B	31	31	262	60	5376	32	0.50	Yes
qwen-next-80b	80	3	262	48	2048	16	0.12	Yes
qwen-2.5-7b	7	7	32	28	3584	28	0.14	No
phi-3.5-mini	3.8	3.8	131	32	3072	32	1.00	No
nemotron-9b	9	9	131	56	4480	40	0.20	Yes
llama-3.1-8b	8	8	131	32	4096	32	0.25	No
granite-3.1-8b	8	8	131	40	4096	32	0.25	No
afm-4.5b	4.5	4.5	65	36	2560	20	0.20	No
apertus-8b	8	8	65	32	4096	32	0.25	No

Table 29: Public architecture and configuration attributes used in Figure 1. Par./Act. = total and active parameters (B), Ctx. = context length (K tokens), Lay./Hid./Hds. = layers, hidden size, attention heads, KV/A = KV-head to attention-head ratio, Hyb. = sparse, sliding-window, MoE, or hybrid attention/state-space components. All models are instruction-tuned variants.

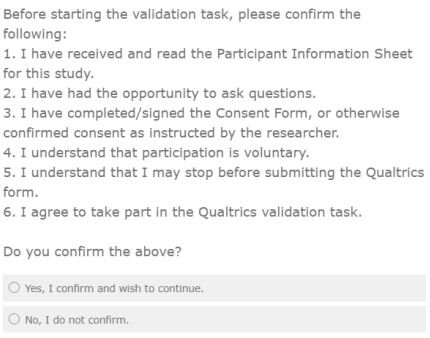
U Human Validation Materials



PARTICIPANT INFORMATION
Study title: A Framework and Benchmark Suite for Information Extraction and Multi-hop Reasoning by LLMs Researcher: [redacted]
REC reference [redacted]

You have been sent the Participant Information Sheet and Consent Form separately. Please read those documents before continuing. This validation task supports [redacted] research project on evaluating specific Retrieval-Augmented Generation-style question-answering systems. You are being invited as a knowledgeable volunteer reviewer to provide an independent human check of benchmark items and LLM-as-judge decisions. You will complete the validation task using a fillable PDF provided by the researcher. After completing the PDF, you will upload it through this Qualtrics form. Participation is voluntary. You may stop before submitting this Qualtrics form.

Fig. 2: Consent confirmation page shown before the validation task.



Before starting the validation task, please confirm the following:
1. I have received and read the Participant Information Sheet for this study.
2. I have had the opportunity to ask questions.
3. I have completed/signed the Consent Form, or otherwise confirmed consent as instructed by the researcher.
4. I understand that participation is voluntary.
5. I understand that I may stop before submitting the Qualtrics form.
6. I agree to take part in the Qualtrics validation task.

Do you confirm the above?

○ Yes, I confirm and wish to continue.
○ No, I do not confirm.

Fig. 3: Participant information page shown before the validation task.

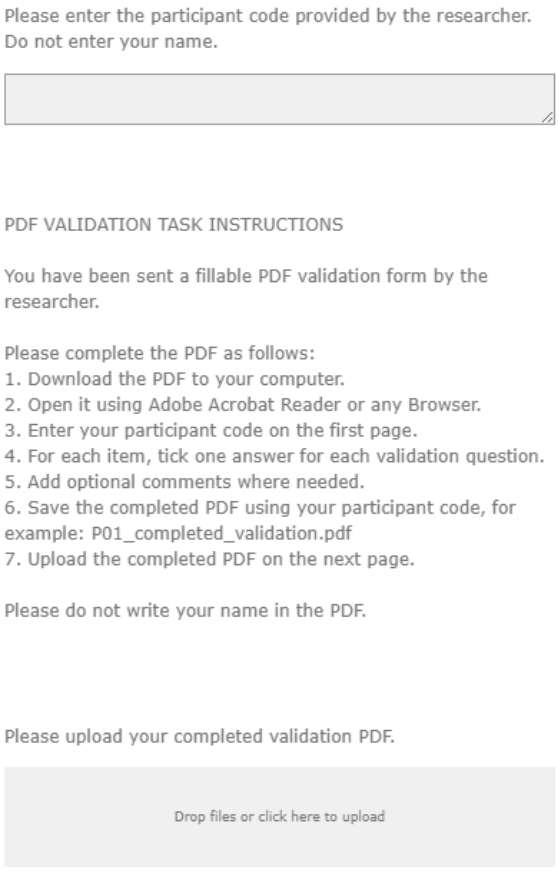
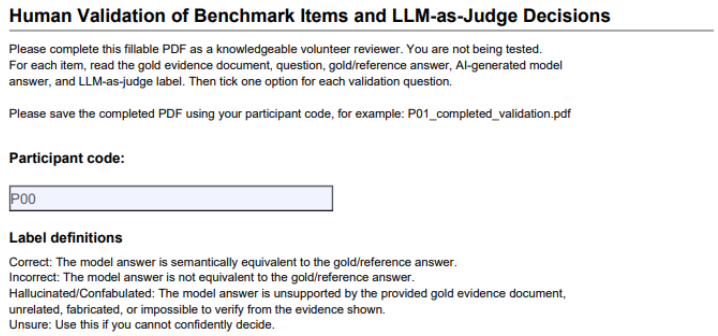


Fig. 4: Qualtrics page for entering the participant code and uploading the completed validation PDF.



Human Validation of Benchmark Items and LLM-as-Judge Decisions

Please complete this fillable PDF as a knowledgeable volunteer reviewer. You are not being tested.
For each item, read the gold evidence document, question, gold/reference answer, AI-generated model answer, and LLM-as-judge label. Then tick one option for each validation question.

Please save the completed PDF using your participant code, for example: P01_completed_validation.pdf

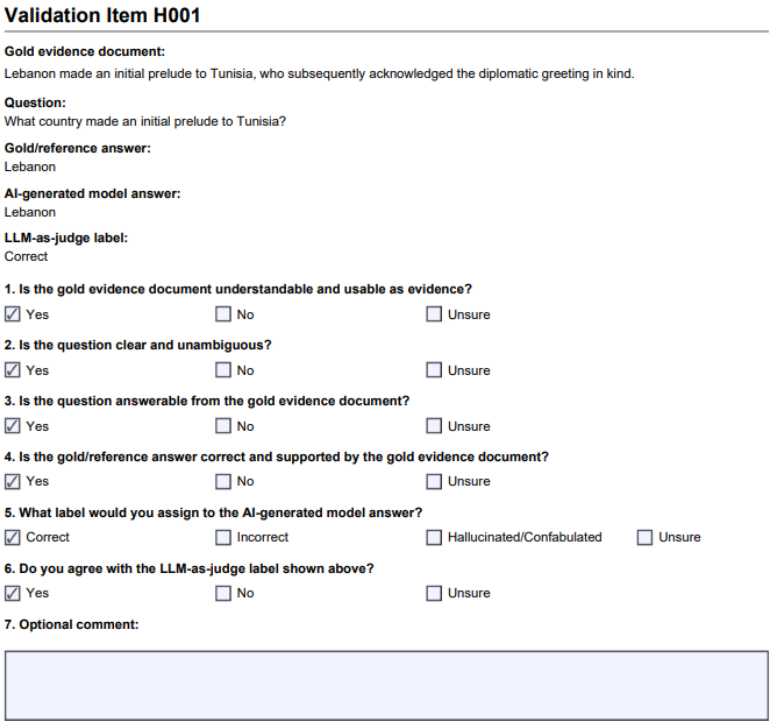
Participant code:

P00

Label definitions

Correct: The model answer is semantically equivalent to the gold/reference answer.
Incorrect: The model answer is not equivalent to the gold/reference answer.
Hallucinated/Confabulated: The model answer is unsupported by the provided gold evidence document, unrelated, fabricated, or impossible to verify from the evidence shown.
Unsure: Use this if you cannot confidently decide.

Fig. 5: Introductory page of the fillable human validation PDF, including participant code and label definitions.



Validation Item H001

Gold evidence document:
Lebanon made an initial prelude to Tunisia, who subsequently acknowledged the diplomatic greeting in kind.

Question:
What country made an initial prelude to Tunisia?

Gold/reference answer:
Lebanon

AI-generated model answer:
Lebanon

LLM-as-judge label:
Correct

1. Is the gold evidence document understandable and usable as evidence?
☑ Yes ☐ No ☐ Unsure

2. Is the question clear and unambiguous?
☑ Yes ☐ No ☐ Unsure

3. Is the question answerable from the gold evidence document?
☑ Yes ☐ No ☐ Unsure

4. Is the gold/reference answer correct and supported by the gold evidence document?
☑ Yes ☐ No ☐ Unsure

5. What label would you assign to the AI-generated model answer?
☑ Correct ☐ Incorrect ☐ Hallucinated/Confabulated ☐ Unsure

6. Do you agree with the LLM-as-judge label shown above?
☑ Yes ☐ No ☐ Unsure

7. Optional comment:

Fig. 6: Example validation item from the fillable PDF (Validated across all hop-depth).

V Non-Gemini Pipeline Regeneration

To test whether benchmark difficulty and the observed degradation pattern depend on Gemini 2.5 Pro specifically, we regenerated a 20% sample of the core Diplomacy benchmark using GPT-5.5 as the sole generator, following the identical construction protocol. We evaluated the same models that we used in Appendix M.

	GPT-5.5-generated		
Model	Hop-1	Hop-4	Δ
Gemini 3.5 Flash	0.89	0.67	−22pp
Claude Sonnet 5	0.71	0.60	−11pp
Claude Haiku 4.5	0.42	0.27	−15pp

Table 30: Frontier model accuracy on a 20% subset regenerated with GPT-5.5,

Accuracy and degradation magnitude are close across generator choice: mean absolute differences are 0.04 at Hop-1 and 0.02 at Hop-4, and all three models show substantial double-digit-point degradation under both generators. Combined with the judge-independence results, this addresses the concern that the observed degradation pattern is a generation artefact specific to the Gemini model family, rather than a property of the reasoning task itself.

W Closed-Book Answerability on Public Benchmarks

To contextualise the closed-book results of our benchmark, we additionally evaluated the same three model families on randomly sampled questions from two publicly available datasets. For NarrativeQA, we sampled 100 questions from distinct narratives; for RGB, we sampled 20 English questions. Each question was presented independently using only its question text, without the associated document, retrieved passages, title, metadata, or reference answers. Responses were evaluated against the provided gold answers using strict semantic-equivalence scoring. As shown in Table 31, all models achieved non-zero closed-book accuracy on both public datasets, whereas none answered a retained instance from our benchmark correctly without context.

X Results

Model	**NarrativeQA** (n = 100)	**RGB** (n = 20)	**Ours (closed-book)**
Gemini 2.5 Flash-Lite	19/100 (19.0%)	14/20 (70.0%)	0.0%
Gemma 4 31B-IT	10/100 (10.0%)	8/20 (40.0%)	0.0%
Qwen-3-Next-80B	12/100 (12.0%)	9/20 (45.0%)	0.0%
Macro average	**13.7%**	**51.7%**	**0.0%**

Table 31: Closed-book parametric answerability on public benchmarks compared with our retained benchmark instances. Models received only the question text. NarrativeQA and RGB therefore serve as public-dataset familiarity controls rather than evaluations under their intended context-provided settings.

Model	ACC (A)	CONFAB↓ (A)	CMD (A)
afm-4.5b	0.150	0.287	0.312
apertus-8b-instruct	0.241	0.473	0.099
gemini-2.0-flash	0.723	0.002	0.992
gemini-2.0-flash-lite	0.521	0.015	0.967
gemini-2.5-flash-lite	0.690	0.000	0.9945
granite-3.1-8b-instruct	0.341	0.126	0.462
llama-3.1-8b-instruct	0.339	0.199	0.273
nvidia-nemotron-nano-9b-v2	0.242	0.285	0.429
phi-3.5-mini-instruct	0.307	0.021	0.953
qwen-2.5-7b-instruct	0.396	0.001	0.997
gemma-4-31b	0.788	0.002	0.98
qwen-3-next-80b	0.628	0.001	0.91

Table 32: Single-hop (n=1) results.

	ACC			CONFAB↓		
Model	A1	A2	A3	A1	A2	A3
afm-4.5b	0.164	0.116	0.170	0.270	0.298	0.292
apertus-8b-instruct	0.236	0.098	0.388	0.548	0.540	0.332
gemini-2.0-flash	0.892	0.680	0.598	0.004	0.002	0.000
gemini-2.0-flash-lite	0.794	0.322	0.446	0.008	0.018	0.020
gemini-2.5-flash-lite	0.896	0.574	0.600	0.000	0.000	0.000
granite-3.1-8b-instruct	0.702	0.148	0.174	0.036	0.172	0.170
llama-3.1-8b-instruct	0.580	0.216	0.220	0.096	0.238	0.264
nvidia-nemotron-nano-9b-v2	0.316	0.174	0.236	0.254	0.308	0.294
phi-3.5-mini-instruct	0.394	0.208	0.320	0.016	0.030	0.018
qwen-2.5-7b-instruct	0.628	0.226	0.334	0.000	0.000	0.002
gemma-4-31b	0.82	0.78	0.75	0.00	0.003	0.001
qwen-3-next-80b	0.78	0.57	0.52	0.006	0.024	0.024

Table 33: Single-hop (n=1) position-resolved ACC and CONFAB for A1–A3. A1: gold document at the beginning, A2: middle, A3: end.

Model	n	ACC (B)	CONFAB↓ (B)	CMD (B)
afm-4.5b	2	0.090	0.340	0.247
afm-4.5b	3	0.051	0.433	0.263
afm-4.5b	4	0.058	0.494	0.298
apertus-8b-instruct	2	0.090	0.628	0.054
apertus-8b-instruct	3	0.051	0.747	0.054
apertus-8b-instruct	4	0.010	0.865	0.054
gemini-2.0-flash	2	0.510	0.003	0.994
gemini-2.0-flash	3	0.436	0.000	0.997
gemini-2.0-flash	4	0.378	0.003	0.987
gemini-2.0-flash-lite	2	0.272	0.003	0.984
gemini-2.0-flash-lite	3	0.218	0.010	0.981
gemini-2.0-flash-lite	4	0.192	0.013	0.987
gemini-2.5-flash-lite	2	0.454	0.003	0.990
gemini-2.5-flash-lite	3	0.376	0.000	0.990
gemini-2.5-flash-lite	4	0.353	0.000	1.000
granite-3.1-8b-instruct	2	0.131	0.077	0.494
granite-3.1-8b-instruct	3	0.119	0.090	0.551
granite-3.1-8b-instruct	4	0.080	0.122	0.455
llama-3.1-8b-instruct	2	0.167	0.304	0.208
llama-3.1-8b-instruct	3	0.099	0.397	0.208
llama-3.1-8b-instruct	4	0.087	0.503	0.272
nvidia-nemotron-nano-9b-v2	2	0.144	0.186	0.587
nvidia-nemotron-nano-9b-v2	3	0.083	0.179	0.673
nvidia-nemotron-nano-9b-v2	4	0.106	0.138	0.750
phi-3.5-mini-instruct	2	0.176	0.032	0.939
phi-3.5-mini-instruct	3	0.179	0.019	0.942
phi-3.5-mini-instruct	4	0.138	0.022	0.923
qwen-2.5-7b-instruct	2	0.167	0.000	1.000
qwen-2.5-7b-instruct	3	0.179	0.003	0.997
qwen-2.5-7b-instruct	4	0.138	0.000	1.000
qwen-3-next-80b	2	0.302	0.051	0.851
qwen-3-next-80b	3	0.249	0.062	0.839
qwen-3-next-80b	4	0.211	0.068	0.855
gemma-4-31b	2	0.625	0.000	0.992
gemma-4-31b	3	0.608	0.000	1.000
gemma-4-31b	4	0.583	0.000	0.983

Table 34: Multi-hop results under a fixed 11-document context budget for hop depth $n \in \{2, 3, 4\}$. Each cell reports A. Lower is better for CONFAB.

		ACC			CONFAB↓		
Model	n	B1	B2	B3	B1	B2	B3
afm-4.5b	2	0.087	0.058	0.125	0.298	0.385	0.337
afm-4.5b	3	0.038	0.058	0.058	0.413	0.413	0.471
afm-4.5b	4	0.048	0.067	0.058	0.481	0.510	0.490
apertus-8b-instruct	2	0.096	0.058	0.115	0.635	0.683	0.567
apertus-8b-instruct	3	0.048	0.038	0.067	0.731	0.769	0.740
apertus-8b-instruct	4	0.010	0.019	0.000	0.875	0.846	0.875
gemini-2.0-flash	2	0.500	0.510	0.519	0.010	0.000	0.000
gemini-2.0-flash	3	0.442	0.404	0.462	0.000	0.000	0.000
gemini-2.0-flash	4	0.404	0.356	0.375	0.010	0.000	0.000
gemini-2.0-flash-lite	2	0.269	0.260	0.288	0.000	0.010	0.000
gemini-2.0-flash-lite	3	0.212	0.212	0.231	0.000	0.000	0.029
gemini-2.0-flash-lite	4	0.183	0.173	0.221	0.019	0.010	0.010
gemini-2.5-flash-lite	2	0.417	0.431	0.515	0.000	0.000	0.010
gemini-2.5-flash-lite	3	0.363	0.410	0.356	0.000	0.000	0.000
gemini-2.5-flash-lite	4	0.353	0.307	0.398	0.000	0.000	0.000
granite-3.1-8b-instruct	2	0.154	0.115	0.125	0.067	0.077	0.087
granite-3.1-8b-instruct	3	0.125	0.115	0.115	0.087	0.106	0.077
granite-3.1-8b-instruct	4	0.067	0.058	0.115	0.106	0.115	0.144
llama-3.1-8b-instruct	2	0.144	0.125	0.231	0.327	0.298	0.288
llama-3.1-8b-instruct	3	0.077	0.077	0.144	0.404	0.404	0.385
llama-3.1-8b-instruct	4	0.087	0.077	0.096	0.500	0.510	0.500
nvidia-nemotron-nano-9b-v2	2	0.144	0.154	0.135	0.212	0.202	0.144
nvidia-nemotron-nano-9b-v2	3	0.077	0.106	0.067	0.173	0.202	0.163
nvidia-nemotron-nano-9b-v2	4	0.096	0.106	0.115	0.154	0.154	0.106
phi-3.5-mini-instruct	2	0.183	0.183	0.163	0.019	0.019	0.058
phi-3.5-mini-instruct	3	0.154	0.173	0.212	0.010	0.019	0.029
phi-3.5-mini-instruct	4	0.163	0.135	0.115	0.019	0.010	0.038
qwen-2.5-7b-instruct	2	0.163	0.135	0.202	0.000	0.000	0.000
qwen-2.5-7b-instruct	3	0.154	0.154	0.231	0.000	0.000	0.010
qwen-2.5-7b-instruct	4	0.144	0.125	0.144	0.000	0.000	0.000
qwen-3-next-80b	2	0.302	0.290	0.314	0.044	0.054	0.056
qwen-3-next-80b	3	0.246	0.240	0.260	0.056	0.064	0.066
qwen-3-next-80b	4	0.224	0.186	0.224	0.064	0.062	0.078
gemma-4-31b	2	0.625	0.625	0.625	0.000	0.000	0.000
gemma-4-31b	3	0.600	0.600	0.625	0.000	0.000	0.000
gemma-4-31b	4	0.625	0.575	0.550	0.000	0.000	0.000

Table 35: Multi-hop formation-resolved results. Rows grouped by model and hop depth $n \in \{2, 3, 4\}$. B1–B3: (1) canonical, (2) reversed, (3) random.

Model	Test	Distracted %	Fabricated %
afm-4.5b	B1	97.88	2.12
afm-4.5b	B2	98.98	1.02
afm-4.5b	B3	97.03	2.97
apertus-8b-instruct	B1	97.22	2.78
apertus-8b-instruct	B2	98.90	1.10
apertus-8b-instruct	B3	99.29	0.71
gemini-2.0-flash	B1	100.00	0.00
gemini-2.0-flash	B2	97.28	2.72
gemini-2.0-flash	B3	94.79	5.21
gemini-2.0-flash-lite	B1	97.98	2.02
gemini-2.0-flash-lite	B2	97.89	2.11
gemini-2.0-flash-lite	B3	98.50	1.50
gemini-2.5-flash-lite	B1	98.08	1.92
gemini-2.5-flash-lite	B2	97.63	2.37
gemini-2.5-flash-lite	B3	99.50	0.50
granite-3.1-8b-instruct	B1	92.48	7.52
granite-3.1-8b-instruct	B2	97.06	2.94
granite-3.1-8b-instruct	B3	97.56	2.44
llama-3.1-8b-instruct	B1	95.68	4.32
llama-3.1-8b-instruct	B2	99.63	0.37
llama-3.1-8b-instruct	B3	99.61	0.39
nvidia-nemotron-nano-9b-v2	B1	99.07	0.93
nvidia-nemotron-nano-9b-v2	B2	98.07	1.93
nvidia-nemotron-nano-9b-v2	B3	98.72	1.28
phi-3.5-mini-instruct	B1	97.30	2.70
phi-3.5-mini-instruct	B2	96.34	3.66
phi-3.5-mini-instruct	B3	97.58	2.42
qwen-2.5-7b-instruct	B1	97.85	2.15
qwen-2.5-7b-instruct	B2	96.64	3.36
qwen-2.5-7b-instruct	B3	96.69	3.31
qwen-3-next-80b	B1	92.64	7.36
qwen-3-next-80b	B2	92.12	7.88
qwen-3-next-80b	B3	90.92	9.08
gemma-4-31b	B1	100.00	0.00
gemma-4-31b	B2	100.00	0.00
gemma-4-31b	B3	100.00	0.00

Table 36: Per-model × per-test error composition among incorrect predictions.

	TestA			
Model	GOLD	P1	P2	P3
afm-4.5b	54.3	75.8	70.4	75.1
apertus-8b-instruct	55.8	76.4	60.3	69.7
gemini-2.0-flash	66.0	80.3	70.7	68.4
gemini-2.0-flash-lite	67.5	88.0	67.8	67.4
gemini-2.5-flash-lite	64.4	88.4	69.5	68.1
granite-3.1-8b-instruct	52.5	80.3	62.5	65.3
llama-3.1-8b-instruct	66.1	81.3	66.1	66.5
nvidia-nemotron-nano-9b-v2	59.2	71.2	66.4	80.1
phi-3.5-mini-instruct	59.2	68.8	67.7	82.3
qwen-2.5-7b-instruct	79.7	87.7	85.5	84.3
qwen-3-next-80b	62.9	78.8	57.0	52.8
gemma-4-31b	80.2	74.2	73.0	62.2
OVERALL	62.6	79.3	69.3	73.9

Table 37: Position-of-distracted (bin presence) for distracted incorrect answers only (percent). For each model, entries report the fraction of distracted cases whose produced answer appears in the gold document (GOLD) and/or in each context third (P1–P3).